

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 Total Marks:50 Annexure A

CONCEPT MAPPING

Code of Conduct:

I (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each

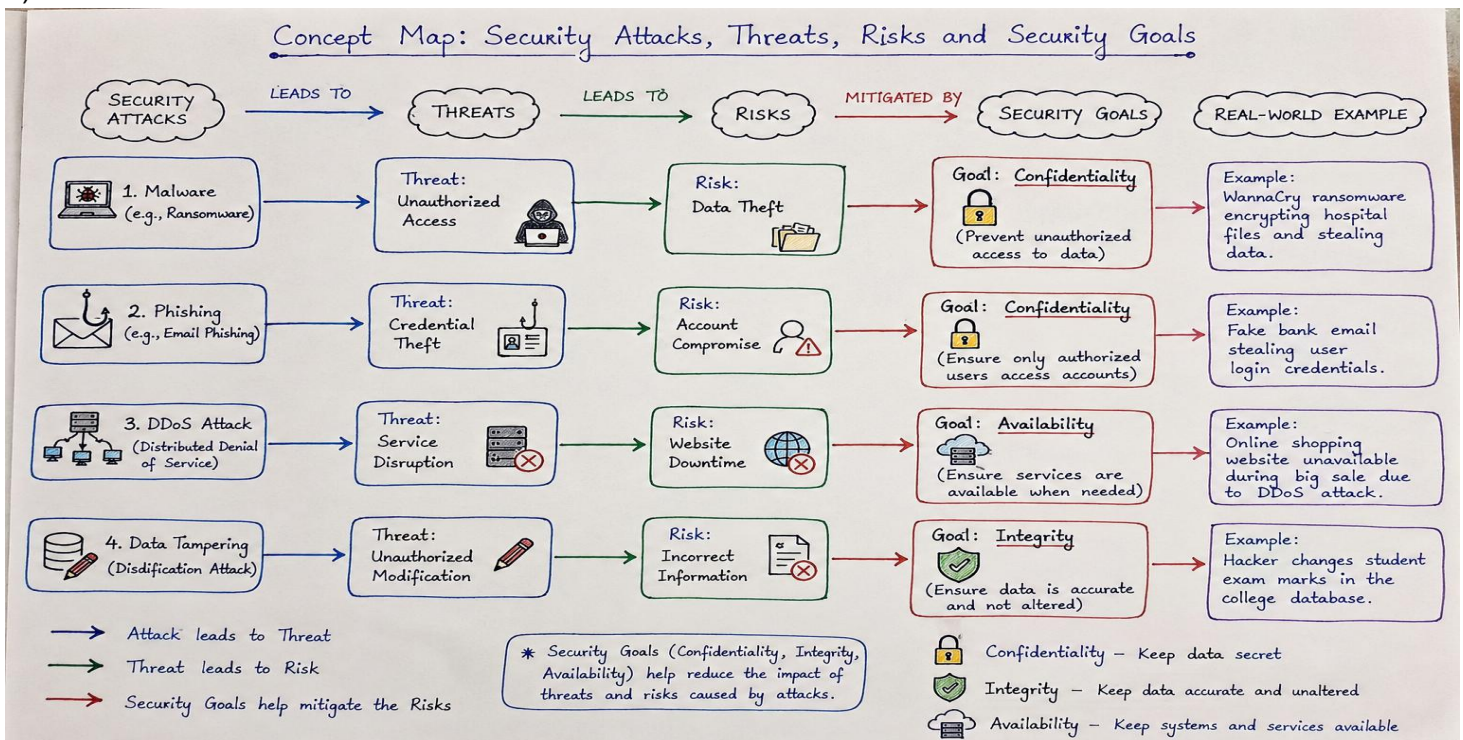
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

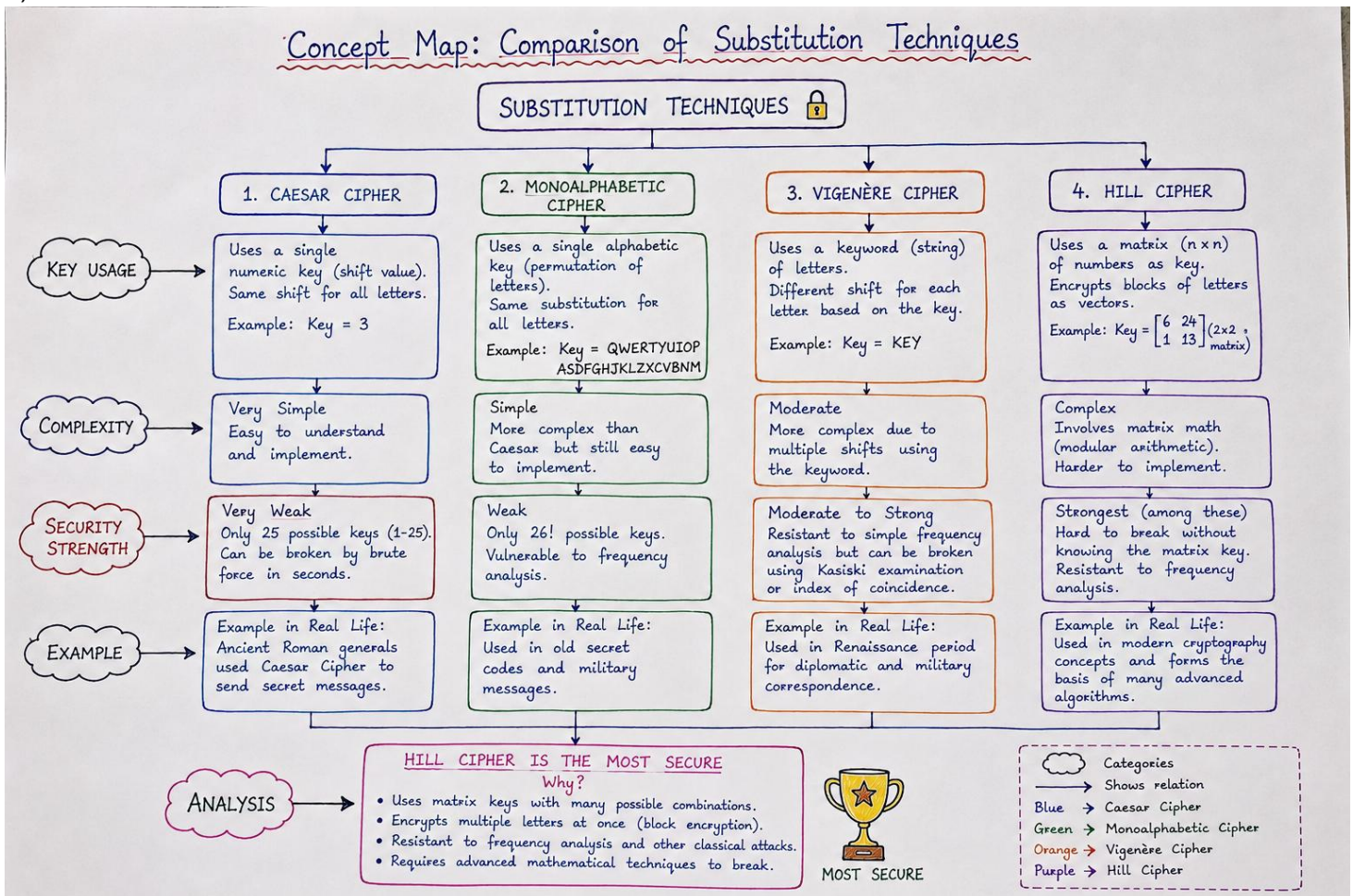
Q. No	Understanding of Concepts (25)	Experimental Design (30)	Use of Tools (20)	Analysis of Results (15)	Documentation (10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

ANSWERS

1)

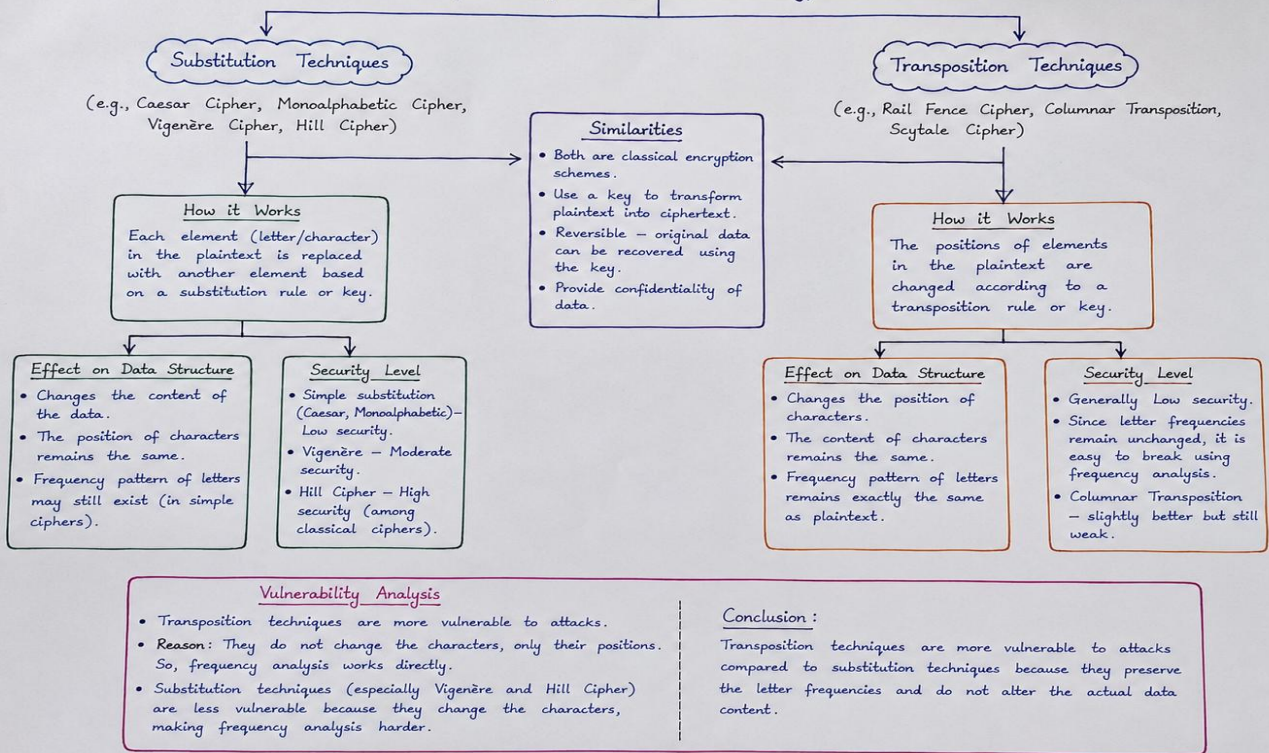


2)



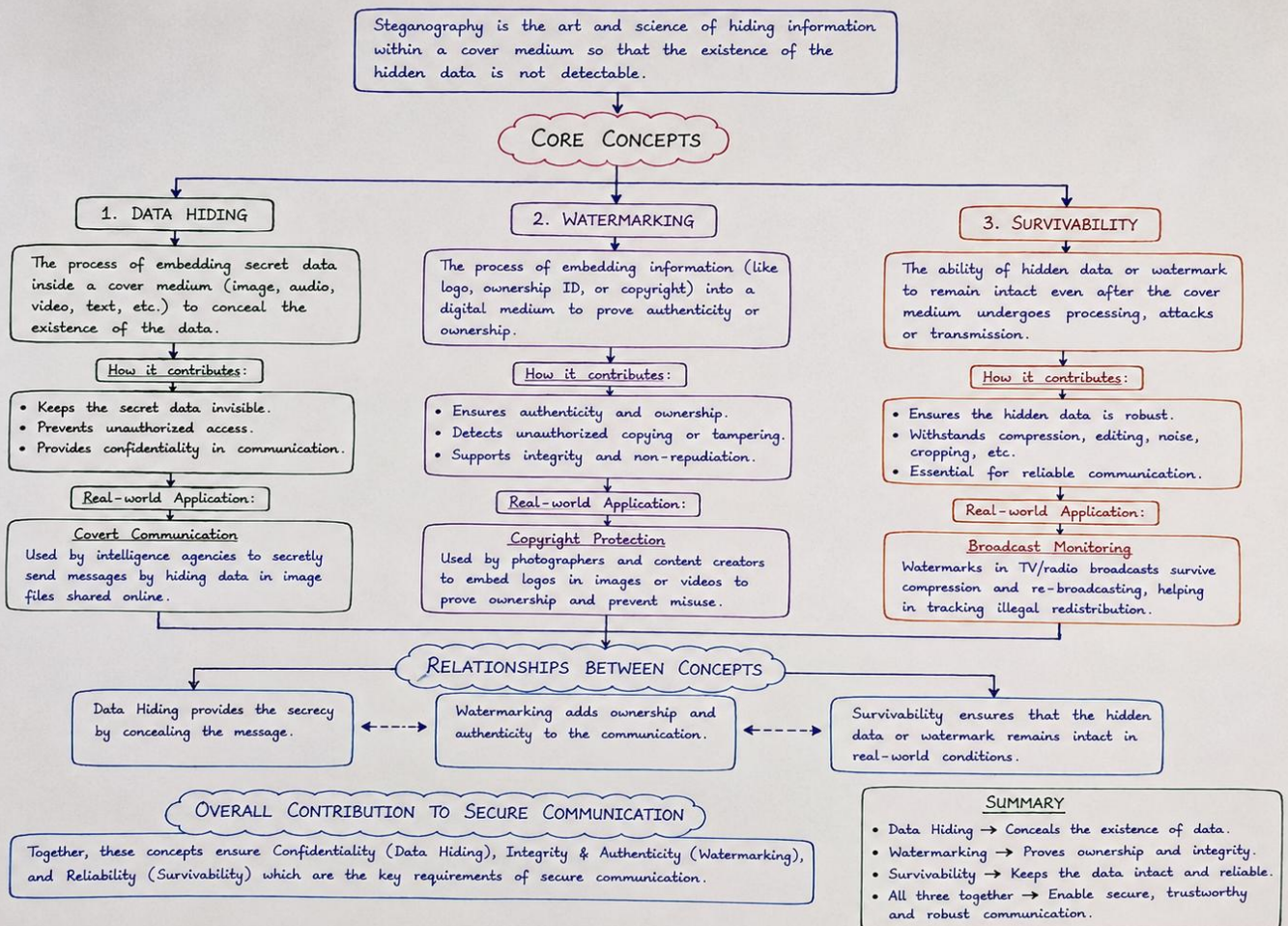
3)

Concept Map: Classical Encryption Schemes



4)

Concept Map: Steganography - Data Hiding, Watermarking and Survivability



5)

Concept Map: Number Theory Concepts used in Cryptography

NUMBER THEORY CONCEPTS IN CRYPTOGRAPHY

1. MODULAR INVERSE

For integers a and m (where $\gcd(a, m) = 1$), the modular inverse of a mod m is a^{-1} such that $a \cdot a^{-1} \equiv 1 \pmod{m}$.

How it connects:

Exists when a and m are coprime. Found using the Extended Euclid Algorithm.

Role in Encryption/Decryption:

Used to find private key (d) from public key (e) in RSA. Decryption: $M \equiv C^d \pmod{n}$ where d is the modular inverse of e mod $\phi(n)$.

Example in Crypto:

In RSA, $d \equiv e^{-1} \pmod{\phi(n)}$ (Private key is modular inverse of public key exponent e).

2. EXTENDED EUCLID ALGORITHM

An algorithm that computes $\gcd(a, b)$ and finds integers x, y such that $ax + by = \gcd(a, b)$.

How it connects:

Used to compute $\gcd(a, m)$. Also finds x such that $ax \equiv 1 \pmod{m}$ when $\gcd(a, m) = 1 \rightarrow (x \text{ is the modular inverse of } a \text{ mod } m)$.

Role in Encryption/Decryption:

Essential for finding modular inverse which is used in key generation and decryption in RSA and other systems.

Example in Crypto:

Find d such that $de \equiv 1 \pmod{\phi(n)}$ using Extended Euclid Algorithm.

3. FERMAT'S LITTLE THEOREM

If p is prime and a is an integer not divisible by p , then $a^{p-1} \equiv 1 \pmod{p}$.

How it connects:

Special case of Euler's Theorem when $n = p$ (prime number). It helps in simplifying exponent calculations in modulo p .

Role in Encryption/Decryption:

Used in primality testing and in simplifying calculations in some cryptographic algorithms. Helps verify correctness in small prime field operations.

Example in Crypto:

If $p = 7$ and $a = 3$ (coprime to 7), then $3^6 \equiv 1 \pmod{7}$.

4. EULER'S PHI FUNCTION

$\phi(n)$ = number of integers in the set $\{1, 2, 3, \dots, n\}$ that are coprime to n .

How it connects:

Used in Euler's Theorem. For prime p , $\phi(p) = p - 1$. For $n = pq$ (product of two primes), $\phi(n) = (p - 1)(q - 1)$.

Role in Encryption/Decryption:

Central to RSA. Used to compute $\phi(n)$ which is needed to find d (private key). Ensures that $ed \equiv 1 \pmod{\phi(n)}$.

Example in Crypto:

For $n = pq$, $\phi(n) = (p - 1)(q - 1)$. Example: $p = 61$, $q = 53 \Rightarrow \phi(n) = 60 \times 52 = 3120$.

5. EULER'S THEOREM

If a and n are coprime (i.e., $\gcd(a, n) = 1$), then $a^{\phi(n)} \equiv 1 \pmod{n}$.

How it connects:

Generalization of Fermat's Little Theorem. When n is prime, $\phi(n) = n - 1$, so it becomes Fermat's Theorem.

Role in Encryption/Decryption:

Foundation of RSA algorithm. Guarantees that if $ed \equiv 1 \pmod{\phi(n)}$, then for any message M coprime to n : $M^{ed} \equiv M \pmod{n}$ (encryption and decryption work).

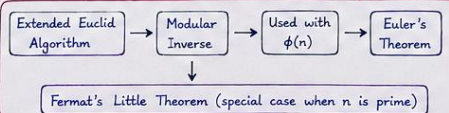
Example in Crypto:

In RSA, since $ed \equiv 1 \pmod{\phi(n)}$, we have: $M^{ed} \equiv M \pmod{n}$.

MATHEMATICAL CONNECTIONS SUMMARY

- Extended Euclid Algorithm $\rightarrow$ used to find Modular Inverse.
- Euler's Phi Function $\rightarrow$ used in Euler's Theorem.
- Fermat's Little Theorem $\rightarrow$ special case of Euler's Theorem when n is prime.
- Modular Inverse + Euler's Theorem $\rightarrow$ foundation of RSA encryption and decryption.

FLOW OF RELATIONSHIPS



OVERALL ROLE IN CRYPTOGRAPHY:

These number theory concepts work together to enable secure key generation, encryption and decryption, especially in algorithms like RSA. They ensure mathematical security, efficiency and correctness.

WHICH CONCEPT IS MOST CRITICAL?

- Euler's Theorem is the most critical concept in public key cryptography (especially RSA) because:
- It provides the mathematical guarantee that encryption and decryption are inverses of each other.
 - It uses $\phi(n)$ (derived from number theory) and modular inverse (found via Extended Euclid Algorithm).
 - Without Euler's Theorem, the correctness of RSA (i.e., $M^{ed} \equiv M \pmod{n}$) cannot be ensured.